

CHE 201 Fall 2019 Quiz 3

Mark Hurley

TOTAL POINTS

60 / 100

QUESTION 1

11 10 / 15

- 0 pts Correct
- 5 pts first one is wrong
- ✓ - 5 pts second one is wrong
- 5 pts third one is wrong

QUESTION 2

22 50 / 85

- 0 pts Correct
- 15 pts mass balance is incorrect
- 20 pts energy balance is incorrect
- 10 pts h1 is incorrect
- ✓ - 10 pts h2 is incorrect
- 10 pts h3 is incorrect
- ✓ - 15 pts m3 is wrong
- 10 Point adjustment

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UIN: _____

QUIZ 3: (100 pts) (Close book/closed notes) Show all your work for full credit

1. (15 points) State whether the following statements are True or False (5 pts each)

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A thermodynamic cycle cannot convert heat completely into work.

TThe change in entropy (ΔS) for a system cannot be less than zero.FA closed system process that violates the 2nd law of thermodynamics must also violate the 1st law of thermodynamics.

2. (85 points) A stream of superheated steam at 3 bar and 280 °C with a mass flow rate of 400 kg/s (stream 1) is mixed with a stream of liquid water at 3 bar and 20 °C (stream 2) in an adiabatic mixing chamber. The resulting mixture exits as a saturated vapor at 3 bar (stream 3). Assume steady state operation and neglect any changes in kinetic and potential energy. Calculate the mass flow rate of the liquid water (stream 2) entering the mixing chamber, in kg/s.

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$$

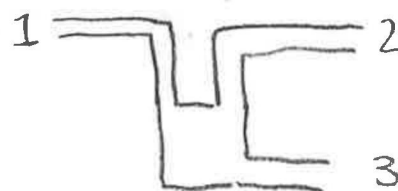
1 joule = 1 N.m 1 N = 1 kg.m/s²

$$\dot{m} = \frac{AV}{v}$$

• no Q • no W • no KE Δ • no PE Δ

$$\dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

$$\dot{m}_1 = 400 \text{ kg/s}$$



Super steam
3 bar 280°C
 $h_1 = 3028.6 \text{ kJ/kg}$ $v_1 = 1.695 \text{ m}^3/\text{kg}$

3 bar 20°C
 $h_2 = 562.2 \text{ kJ/kg}$ $v_2 = 0.0058 \text{ m}^3/\text{kg}$

Sat vapor
3 bar
 $h_{3g} = 2725.3 \text{ kJ/kg}$

$$v_2 = 0.0058 \text{ m}^3/\text{kg}$$

$$\sum \dot{m}_i + \sum \dot{m}_E = 0$$

$$\dot{m}_3 = \dot{m}_1 + \dot{m}_2$$

$$\dot{m}_1 = \dot{m}_3 - \dot{m}_2$$

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$$\dot{m}_1 h_1 = \frac{\dot{m}_3 h_3}{\dot{m}_2 h_2}$$

